Auto-Curriculum

Letting a network choose its own blur and contrast curriculum

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CMPE492 Senior Project Proposal · October 2026

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1. Motivation and background

Infants start with poor acuity (about 20/400) and low contrast sensitivity. Recent work argues this early degradation is adaptive. CNNs trained blurry-to-sharp form broader receptive fields (Vogelsang et al. 2018), and high-to-low internal noise improves robustness (Arslan, Jang, Sinha 2024).

The evidence is mixed. For objects, a fixed blurry-to-clear sequence loses its blur robustness (Jang & Tong 2021), while mixing blur throughout training helps (Jang & Tong 2024). Gains may also come from exposure to blur, not from order (Katzhendler & Weinshall 2019). All of these schedules are designed by hand.

On the curriculum side, Graves et al. 2017 let a bandit pick training tasks by how fast the network is learning, and Portelas et al. 2019 extend this to continuous spaces. EfficientTrain uses a low-frequency-first schedule, found offline to save compute. Wu et al. 2021 and Hacohen & Weinshall 2019 warn that curriculum gains are often small.

2. Research question and hypotheses

Question: if the network chooses the blur and contrast of its own training images, based only on how fast it is learning, does it start blurry and faint like an infant, and does that make it more robust?

- **H1, an infant-like order emerges:** early training favours blurry, low-contrast images, and later training moves toward sharp, full-contrast images.
- **H2, robustness follows:** this self-chosen curriculum improves robustness to image corruptions not seen in training, and increases shape bias, compared with controls that see the same images in a different order.

Curriculum gains often shrink with long training (Wu et al. 2021). We therefore test both a short and a full training budget.

3. Approach

We train a standard image classifier, for example a ResNet-18 on CIFAR-10. Before each training step, a teacher chooses how blurry and how low-contrast the images will be. The teacher is a multi-armed bandit, as in Graves et al. 2017: each arm is one blur and contrast level, and the bandit is rewarded when the network learns quickly at that level.

Robustness and shape bias are not part of the reward. We measure them only after training, so the teacher cannot be steered toward them. To make sure the result does not depend on one design choice, we repeat it with a second learning-progress signal and with a continuous version of the teacher (ALP-GMM).

4. Experiments

We compare the bandit with controls that separate the order of the images from simple exposure to blur and low contrast. Each condition runs with several random seeds and the same compute.

CONDITION	WHAT IT TESTS
Bandit teacher	Main method
Bandit teacher, alternative progress signal	Is the order caused by the reward signal?
Sharp images only	Standard baseline
Random level for each batch	Exposure without order
Bandit schedule, shuffled in time	Order versus exposure
Hand-designed blurry to sharp	Does adapting beat a fixed schedule?
Sharp first, then blurry	Does the direction matter?

Measures: clean accuracy; robustness on CIFAR-10-C, reported separately for corruptions similar to our training transforms (blur, contrast) and for unseen ones (noise, weather, digital); shape bias on cue-conflict images (Geirhos et al. 2019); and the curriculum the bandit chose over time.

5. Contribution

- **Compared with EfficientTrain:** they fix a low-frequency-first schedule offline to save compute and do not test robustness. Our schedule emerges during training from learning progress, and we test robustness and shape bias.
- **Compared with developmental schedules** (Vogelsang; Arslan et al.; Jang & Tong): they impose the order by hand. We ask whether the network chooses it. Our teacher can also return to blurry images, which may avoid the forgetting seen with fixed blurry-to-clear schedules.
- **Compared with Graves and Portelas:** same teacher idea. In the work we reviewed, it has not been applied to image degradation or to robustness.

6. Work plan

1. **Set-up:** fix the dataset, the model and the blur and contrast levels. Build the training pipeline and the robustness and shape-bias evaluations.
2. **Teacher:** implement the bandit teacher with the learning-progress reward. Check that a teacher paired with a network that does not learn shows no order.
3. **Experiments:** run all conditions with several seeds, first with a short training budget, then with a full one.
4. **Robustness checks:** repeat the main comparison with the alternative progress signal and the continuous teacher.
5. **Analysis and report:** describe the curricula the teacher chose, compare robustness and shape bias across conditions, and write the final report.

7. Open design decisions

- The range and number of blur and contrast levels.
- The robustness benchmark: CIFAR-10-C, or the noise tests used by Arslan et al. 2024.
- How to measure shape bias on CIFAR: a custom cue-conflict set, or a move to an ImageNet subset.